

EDUCATION

- **Technische Universität Darmstadt** Darmstadt, Germany
Semester Exchange Student; CGPA: 8.44 Oct 2025 – March 2026
 - **Relevant Coursework:** Artificial Intelligence, Computer Vision, Probabilistic Graphical Models, Large Language Models, Scientific Computing
- **Indian Institute of Technology, Mandi** Mandi, Himachal Pradesh, India
B.Tech in Bioengineering; CGPA: 8.02/10 Aug 2023 – May 2027
 - **Relevant Coursework:** Data Structures & Algorithms, Machine Learning, Introduction to Python and Data Science, Probability and Statistics

EXPERIENCE

- **HIMPACT LAB, IIT Mandi** Mandi, Himachal Pradesh, India
Frontend Developer Intern Aug 2024 – Dec 2024
 - **UI Development:** Built the first production-ready UI for Incline using Django templates, HTML/CSS, and vanilla JS; ensured responsive layouts across devices.
 - **Backend Integration:** Integrated Leaflet.js-based interactive maps into the Django backend; enabled real-time server-side plotting and dynamic data retrieval.
 - **Project Delivery:** Handed off production-ready version to the next developer; live at incline.iitmandi.ac.in.

TECHNICAL SKILLS

- **Languages:** Python, C++, JavaScript, Perl
- **Frameworks:** React, ROS, TensorFlow, Django, Streamlit
- **Tools:** Git, Linux, Docker, SolidWorks, Raspberry Pi, Arduino
- **Concepts:** Sensor Fusion, Kalman Filtering, Data Preprocessing, Control Systems

PROJECTS

- **Predicting Protein Stability Changes ($\Delta\Delta G$)** [Live](#)
Machine Learning, Siamese GNN, ESM-2, Bayesian Networks Feb 2026 – Present
 - **Architecture Design:** Engineered a 1D-3D latent fusion architecture combining a Siamese Graph Neural Network (GNN) with a 150-million parameter Protein Language Model (ESM-2).
 - **Predictive Performance:** Achieved a Pearson Correlation of 0.7266 and Mean Absolute Error of 0.7472 kcal/mol, significantly surpassing traditional biophysical force fields.
 - **Causal Inference:** Constructed a Hybrid Bayesian Network to map explicit biological causality, mathematically quantifying the vulnerability of protein surfaces to polarity shifts.
- **Vaya – Your Local Healthcare Connection** [GitHub](#)
Python, Django, Django Channels, Groq API, RAG Aug 2025 – Present
 - **Full-stack Platform:** Designed doctor discovery, appointment booking, and report handling system with secure authentication.
 - **AI RAG System:** Implemented a two-step retrieval-augmented generation pipeline reducing hallucinations and improving accuracy.
 - **Real-time Messaging:** Built encrypted, real-time patient-doctor chat with Django Channels.
- **AquaSweep – Underwater Rover** [GitHub](#)
Python, OpenCV, Flask, ROS, Arduino, Raspberry Pi Jan 2025 – Jun 2025
 - **Live Streaming:** Engineered MJPEG camera pipeline (20 FPS, ~200ms latency) using Flask sockets.
 - **Edge AI Detection:** Built OpenCV pipelines for algae and motion detection with 93% lab accuracy.
 - **Depth Stabilization:** Controlled 8 thrusters via ROS-Arduino; fused IMU and Bar30 data with Kalman filtering ($\pm 3\text{cm}$ error).

LEADERSHIP & ACHIEVEMENTS

- Selected for semester exchange at TU Darmstadt (Top 450+ batch merit).
- Hospitality Head, Exodia: Led 30-member team; managed 1000+ participants and logistics.
- Robotics Club, IIT Mandi: Mentored juniors on robotics design and automation projects.
- Secured Top 5 rank in Arduino Hackathon during intra-college techfest.